

# Data-driven Data Augmentation for Motor Imagery Brain-Computer Interface

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**Abstract**—When utilizing deep learning (DL) for the brain-computer interface (BCI), it suffers from degradation of classification performance due to the small quantity of electroencephalography (EEG) training data. Typically, artifacts such as blinking and a falling-off in user's attention caused by a long experiment period have a great effect on the small quantity and the low quality of the EEG training data. To address this problem, in this work, we introduce a novel classification method based on data augmentation method and DL framework for motor imagery (MI) EEGs classification. The proposed framework generates input EEG images with distinct MI features from a small amount of EEG training data. Experimental results using a public BCI dataset show that the proposed method outperforms the predecessors which use a simple DL model.

**Keywords**—Brain-computer interface, motor imagery, ensemble empirical mode decomposition, filter bank common spatial pattern, data augmentation, convolutional neural network

## I. INTRODUCTION

Brain-computer interface (BCI) which is a technology to control various external applications by decoding brain signals with the user's intention has received attention from a number of the researcher in company with the development of deep learning (DL). The state-of-the-art DL models have greatly contributed to the improvement of the classification performance in BCI [1], [2]. However, in terms of the amount of electroencephalography (EEG) training data, BCI is still suffering compared to other studies. Typically, the expensive cost for a number of subjects and subject's fatigue caused by the long-term experiment are exemplified. Also, artifacts such as eyes blinking and muscle activity interfere with the construction of high-quality EEG training data.

To tackle some of the above-mentioned problems, data augmentation methods-based a variety of DL models have been proposed [3-5]. In [3], the authors introduced several data augmentation methods such as multiplication, flip, data skew, etc. The methods have solved imbalance and limits on the size of datasets and improved classification performance. In another study [5], Zhang *et al.* proposed a novel DL framework applied a new data augmentation strategy by extracting and combining intrinsic mode function (IMF) using empirical mode decomposition (EMD) [6]. However, according to [7], one of the major drawbacks of the EMD is mode mixing which is denoted as any IMF with very similar oscillations in disparate mode. Therefore, the artificial EEG training data generated by the data augmentation method applied EMD has limitations in an increasing volume of the data while maintaining high

quality. In this article, to deal with the aforementioned problem, we suggest a novel MI classification method based on data augmentation method designed using ensemble empirical mode decomposition (EEMD) present by Wu and Huang [7]. In order to validate the classification performance of our method, among various BCI paradigms, we utilize the motor imagery (MI) paradigm, which consists of left hand MI and right hand MI. Though benchmark MI datasets, we first multiply the increasing ratio of the original data from 2 times to 5 times, and then generate an input image with a distinct feature of each hand MI by filter bank common spatial pattern (FBCSP) [8]. After that, as a classifier model, a shallow one-dimensional convolutional neural network (1D CNN) was utilized. According to experimental results evaluated from 5-fold cross-validation for each subject, the proposed method with the data augmentation method based on EEMD was demonstrated to be better than the method based on EMD.

## II. METHODOLOGY

### A. Ensemble empirical mode decomposition (EEMD)

EEMD is defined as a noise-assisted data analysis method to overcome the mode mixing problem in EMD [7]. The IMFs extracted in EEMD are calculated as the mean of the IMFs obtained by EMD on contaminated signals by adding Gaussian white noise. The formula of the EEMD can be explained by the following steps. First, Gaussian white noise is added to the original signal.

$$x_k(t) = x(t) + w_k(t) \quad (1)$$

where  $x(t)$  is an original signal,  $w_k(t)$  is  $k$ th Gaussian white noise. Then, the corrupted signals decompose into the IMFs using EMD.

$$x_k(t) = \sum_{n=1}^{N-1} IMF_n^{(k)}(t) + r_n^{(k)}(t) \quad (2)$$

where  $N-1$  is the number of IMFs,  $IMF_n^{(k)}$  is the  $n$ th IMF and  $r_n^{(k)}$  is the residual. The IMFs of EEMD are calculated as

$$IMF_n = \frac{1}{K} \sum_{k=1}^K IMF_n^{(k)}(t) \quad (3)$$

EEMD with an ensemble approach has become an upgraded version of EMD for nonstationary and nonlinear time series analysis.

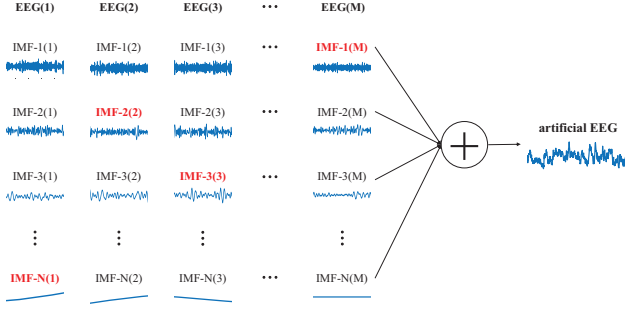


Fig. 1. Illustration of the proposed data augmentation method.

### B. Filterbank common spatial pattern (FBCSP)

The common spatial pattern (CSP) approach is a feature extraction method widely used in two-class MI task classification. The purpose of CSP is to search a set of linear transformations such as the spatial patterns which maximize the variance for one class and meanwhile minimize the variance for the other class. The spatial filtered signal  $\mathbf{Z} \in \mathbb{R}^{c \times n}$ , in which single-trial EEG signal  $\mathbf{X} \in \mathbb{R}^{c \times n}$  is projected by the CSP projection matrix  $\mathbf{W} \in \mathbb{R}^{c \times c}$ , is given as

$$\mathbf{Z} = \mathbf{W}\mathbf{X} \quad (4)$$

where  $c$  is the number of EEG channels and  $n$  is the number of EEG samples. The rows of  $\mathbf{W}$  represent the spatial filters, which are orthogonal to each other. Generally, several  $\epsilon$  spatial filters are chosen among whole filters [9]. The spatial filtered signal obtained by the  $\epsilon$  first and last rows of spatial filters is extracted as inputs to a classifier.

$$f_k = \log \left( \frac{\text{var}(\mathbf{Z}_k)}{\sum_{k=1}^{2\epsilon} \text{var}(\mathbf{Z}_k)} \right) \quad (5)$$

However, there is a problem that CSP needs to manually set a broad or subject-specific frequency range. Thus, FBCSP has been proposed to address this problem [8]. The FBCSP is extracted the CSP features based on the EEG filtered by the filter bank. The filter bank was comprised of Chebyshev Type II band-pass filters to decompose the EEG signal into multiple frequency bands. For  $i$ th frequency band filtered signal, the formula of the features yielded through the CSP is as given by

$$f_{i,k} = \log \left( \frac{\text{var}(\mathbf{Z}_{i,k})}{\sum_{k=1}^{2\epsilon} \text{var}(\mathbf{Z}_{i,k})} \right) \quad (6)$$

### C. The proposed method based on data augmentation

As mentioned in Section II-A, the IMFs extracted by EEMD were employed to make the quality artificial EEG data. Firstly, the IMFs separated from the original EEG data and the trials were randomly selected for the same channel number, and then applied in one new artificial EEG data. Here, the number of IMFs and trials used in artificial EEG data was set as 15, and additional IMFs with zero value was inserted to equalize the number of IMFs for all original EEG signals. The architecture of the data augmentation method is illustrated in Fig. 1. For the given parameters, each  $M$  and  $N$  denote the number of trials and IMFs.

Next, the FBCSP was performed to create the input image with a clear MI feature of the DL model. Fig. 2 represents the averaged input image for all trials of subject 8. In this figure,

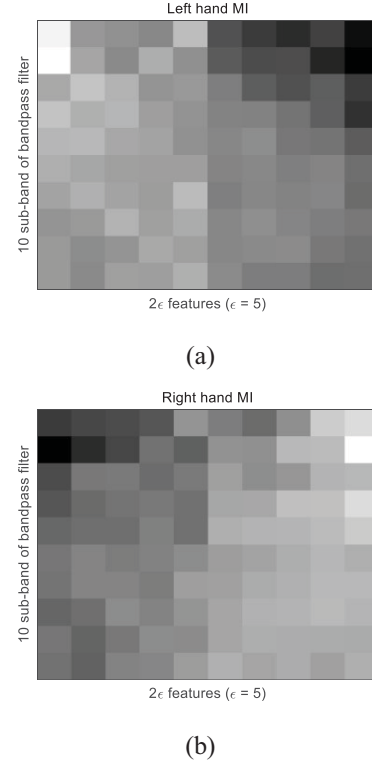


Fig. 2. Averaged input Images for all trials in each MI task (a) Left hand MI (b) Right hand MI. As can be seen, the contrasting features for each MI task emerging in mu band (8-13 Hz) are obviously indicated. These observations demonstrate that the MI task feature has learned by the proposed method is useful.

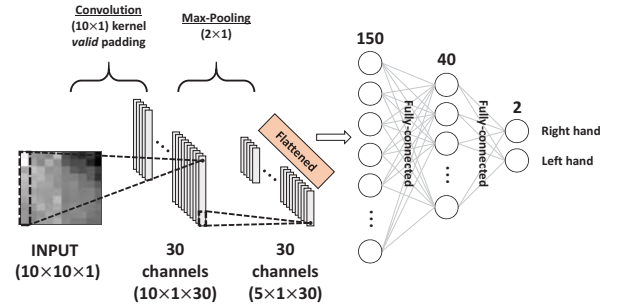


Fig. 3. The structure of proposed 1D CNN.

the y-axis indicates that the lower and upper cut-off frequencies of the  $i$ -th ( $i = 1, \dots, 10$ ) sub-band were set to  $i \times 4$  Hz and 46 Hz with 6 Hz interval and 2 Hz overlapping, respectively. The x-axis is the number of spatial filter pairs, which are set to 5 in this study. The MI features are shown strongly in mu band (8-13 Hz) compared to other frequency bands. For example, as shown in figures 2(a) and 2(b), the pixel values in the upper position of the left and right sides of the input images show the opposite appearance.

As the classifier model, one-dimensional CNN (1D CNN) was utilized. This model is multi-layer neural networks with one 1D convolution-max pooling layer pairs and two fully connected layers, and the structure is shown in Fig. 3. The 1D convolutional layer is activated by the rectified linear unit (ReLU) function. The parameters of the proposed model were trained

TABLE I. BCI COMPETITION IV DATASET 2A AVERAGE ACCURACY (%) AND STANDARD DEVIATION (STD. DEV) RESULTS FOR ORIGINAL AND DATA AUGMENTED DATASET.

| Subjects                     | Average Accuracy % |             |             |             |             |             |             |             |              |
|------------------------------|--------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|
|                              | Original           | EMD         |             |             |             | EEMD        |             |             |              |
|                              |                    | 2 × Dataset | 3 × Dataset | 4 × Dataset | 5 × Dataset | 2 × Dataset | 3 × Dataset | 4 × Dataset | 5 × Dataset  |
| A01                          | 95.14              | 93.06       | 93.29       | 94.45       | 94.65       | 95.32       | 96.64       | 95.75       | 96.95        |
| A02                          | 67.05              | 76.57       | 79.86       | 85.42       | 86.32       | 75.19       | 81.37       | 83.44       | 84.31        |
| A03                          | 92.68              | 88.02       | 90.40       | 92.79       | 88.47       | 92.19       | 93.86       | 91.58       | 92.78        |
| A04                          | 67.72              | 81.44       | 85.30       | 89.49       | 89.16       | 83.85       | 88.09       | 88.46       | 88.75        |
| A05                          | 72.54              | 76.36       | 83.03       | 82.46       | 81.95       | 75.53       | 79.38       | 85.85       | 87.43        |
| A06                          | 68.13              | 72.57       | 80.90       | 86.64       | 82.92       | 70.38       | 77.86       | 80.47       | 82.01        |
| A07                          | 91.36              | 81.95       | 84.96       | 85.89       | 86.94       | 89.58       | 91.55       | 91.14       | 92.43        |
| A08                          | 98.25              | 94.62       | 96.19       | 97.48       | 97.36       | 96.87       | 97.69       | 98.18       | 98.61        |
| A09                          | 94.47              | 94.78       | 96.53       | 96.18       | 96.46       | 96.18       | 96.64       | 97.66       | 97.92        |
| <b>Average</b>               | 83.04              | 84.37       | 87.83       | 90.09       | 89.36       | 86.12       | 89.23       | 90.28       | <b>91.24</b> |
| <b>Std. Dev.<sup>a</sup></b> | 12.88              | 8.03        | 6.07        | 5.03        | 5.31        | 9.63        | 7.44        | 5.91        | <b>5.67</b>  |

<sup>a</sup> Standard Deviation

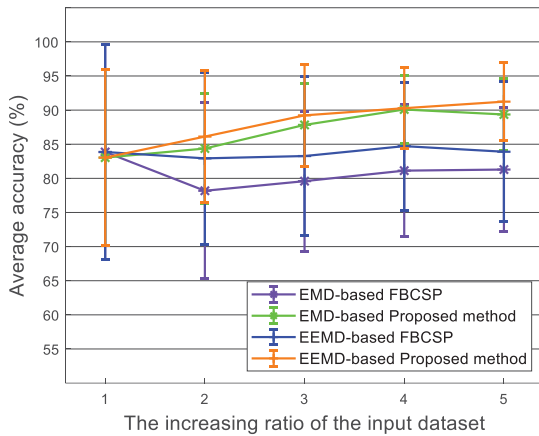


Fig. 4. The average accuracy across all subjects obtained by the four methods using different the increasing ratio of the input dataset. The error bars indicate standard deviations.

using Adam optimizer. Also, we use 50 as the mini-batch size and the cross-entropy function as the loss function.

### III. RESULTS

#### A. Dataset

We adopt a benchmark dataset, i.e., BCI competition IV dataset 2a, to estimate the proposed method. There are nine healthy subjects for four different MI tasks such as left hand, right hand, feet, and tongue in the dataset. Among them, we select both hands MI EEG datasets. Each subject carried out two sessions on different days. Here, each session consists of 6 runs and one run includes 48 trials. The EEG data was recorded from 25 electrodes, i.e., 22 EEG electrodes and 3 electrooculography (EOG) electrodes, and sampled at 250 Hz. Also, the data was extracted the frequency bands between 0.5 Hz and 100 Hz using a bandpass filter and removed 50 Hz

using a notch filter. More detailed information was reported in [9].

#### B. Experimental results

The proposed method is compared with the previous methods using 5-fold cross-validation. Table I. shows the comparison of the subject-specific average accuracy and standard deviation obtained by data augmentation methods based on the EMD and the EEMD. According to the table, the best average accuracy across all subjects for the EMD and the EEMD is 90.09% and 91.24%, respectively. For both two methods, we were able to confirm that the classification performance has improved with the data augmentation of the original training dataset. However, the optimal increasing ratio for each method is different. For instance, the EMD has shown the best average accuracy at the 4 times magnifications, while the best average accuracy of the EEMD has indicated at the 5 times magnifications.

Fig. 4 illustrates the average accuracies for the proposed method and the other method such as FBCSP at various increasing ratios of the input dataset. In the FBCSP procedure, linear discriminant analysis (LDA) was performed as a classifier model. As shown, our method yields better average accuracy at more than 2 times magnifications compared to other methods. For the performance using FBCSP, the average accuracy has somewhat deteriorated as the amount of data increased. These results demonstrate that our method could be a promising method for developing a high-performance approach for both hand MI classification.

All the experiments were conducted in MATLAB and Python under the framework of TensorFlow [10] environment on an Intel Core i7-6850K CPU, a NVIDIA GeForce GTX 1080 TI GPU and 32 GB RAM.

### IV. CONCLUSIONS

This paper introduces a novel classification methodology including the proposed data augmentation method and DL

framework to classify the left and right hand MI. To overcome the drawback of the EMD-based data augmentation method, the quality of EEG training data can be improved by designing the EEMD-based data augmentation method. In addition, the proposed method devised using FBCSP and 1D CNN has validated enhanced left and right MI tasks classification performance through the experimental results. Future works include exploring more expressive deep learning model as well as a qualitative data augmentation approach to improve the performance of MI tasks classification.

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